

Endogenous Population Scale in the Existing Housing–Fertility Model

Research update

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Abstract

The paper’s lifecycle model determines fertility, tenure, and housing choices for a stationary population. This update adds one feedback: children become future households, and those households enter the housing market. A simple two-generation model shows how fertility and housing-market clearing jointly determine the long-run housing cost and population. The quantitative exercise then adds the population closure around the existing household model, without changing its preferences, states, or policies. It shows how the saved circulated one-shot estimate can be interpreted as an open steady state, where a closed steady state lies, and how allowing population to adjust changes the incidence of housing and credit policies. A historical transition first requires an empirical bridge from mature children to household formation and, beyond that, a forward-looking asset-price path. It is therefore kept separate from the results reported here.

1 The mechanism

Children raise the demand for space at the same ages when households have low liquid wealth and limited access to ownership. Large rental units are scarce, and buying a family-sized home requires a down payment. Older households, by contrast, own much of the large-housing stock and may retain it after their children leave because moving is costly and housing is part of the estate they plan to leave. The allocation of housing across generations can therefore raise the effective cost of a child even when the aggregate stock is large.

The feedback has two directions. First, a higher housing cost or a tighter access constraint reduces fertility. Second, children eventually form the next generation of households, so fertility changes future housing demand. The first direction is a household mechanism; the second is a demographic equilibrium condition. Combining them changes both the transition and the long-run incidence of housing policy.

These links have close precedents. Demographic effects on housing demand go back at least to Mankiw and Weil (1989), and recent evidence relates birth-cohort size to housing prices through entry into and exit from homeownership (Francke and Korevaar, 2025). Housing prices and mortgage access also affect childbearing (Lovenheim and Mumford, 2013; Detting and Kearney, 2014; Hacamo, 2021). Malmberg (2012) studies the two-way relation between cohort housing pressure and fertility, while Lee and Suh (2025) analyze a low-fertility housing transition. Our question is narrower: how does the incidence of housing policy change when tenure constraints and old-age retention affect fertility, and births in turn determine the number of future households?

The current paper already writes a stationary entry condition, while its household computations describe a population normalized to unit size. Let S be the number of adult households in the market. For any housing cost p , the household model tells us four things: the distribution of households, housing demand per household $\bar{h}(p)$, the number of new households required to replace those that exit $E(p)$, and the number produced by children born inside the model $B(p)$. The objects $E(p)$ and $B(p)$ are flows per adult household and model period.

To allow for immigration, let M be the aggregate number of households arriving from outside in the same period and let ρ be the fraction of locally born households that remain in the market. A stationary population and a clearing housing market then require

$$SE(p) = M + \rho SB(p), \quad S\bar{h}(p) = H^s(p). \quad (1)$$

The first equation says that local children and outside entrants together replace exiting households. The second says that total housing demand equals housing supply. These two equations determine population S and the housing cost p without changing the household problem.

For a stationary comparison, this is the entire population extension. A transition additionally requires the age distribution and the children who have already been born but have not yet formed households. Those objects are introduced below. Outside and locally born entrants are initially assigned the same entry-state distribution as in the current model; allowing their wealth or income distributions to differ is a later extension.

The next section presents the same mechanism with young and old households. Section 3 then maps it into the current paper.

2 Two-generation theory

2.1 Environment

Young households. At date t , there are Y_t young households. Each has income y and chooses consumption c , housing for the adults h , and the number of children n . Preferences are

$$u(c, h, n) = c + \alpha \log h + \theta \log n, \quad \alpha > 0, \quad \theta > 0. \quad (2)$$

The parameter α governs the demand for adult housing and θ governs the demand for children. Quasi-linearity makes both demands available in closed form. We assume $y > \alpha + \theta$, so consumption is positive at the interior solution.

Child costs and young access. Each child requires $q \geq 0$ units of other goods and $a > 0$ units of housing. One unit of housing costs p . Obtaining family space also entails an access cost $\omega \geq 0$, which represents a small rental market, a down-payment constraint, or another tenure friction. The cost of a child is therefore $q + a(p + \omega)$.

Old households. There are O_t old households. They no longer choose fertility, and their housing demand is

$$h_O(p) = \frac{\bar{h}_O}{p}, \quad \bar{h}_O > 0. \quad (3)$$

The parameter $\bar{h}_O$ summarizes old households' desire to retain housing. A policy that encourages downsizing reduces $\bar{h}_O$. In the full model, housing retention follows from moving costs and the value of leaving a house to one's children rather than from this demand schedule.

Housing. Aggregate housing services are supplied according to

$$H^s(p) = \bar{H}p^\varepsilon, \quad \bar{H} > 0, \quad \varepsilon \geq 0. \quad (4)$$

The fixed-stock case is $\varepsilon = 0$.

Demography. The number $\bar{n} > 0$ is replacement fertility. After survival and household formation, $\bar{n}$ children produce one new young household.

In this static section, p is the current cost of one unit of housing services. The lifecycle model distinguishes that current rent, denoted r_t , from the asset price P_t that capitalizes future rents. They move proportionally only under a constant user cost and a stationary price path.

2.2 The young household

Taking p as given, a young household solves

$$\max_{c,h,n} c + \alpha \log h + \theta \log n \quad \text{subject to} \quad c + ph + [q + a(p + \omega)]n = y. \quad (5)$$

At an interior solution, the household chooses

$$n(p) = \frac{\theta}{q + a(p + \omega)}, \quad h_Y(p) = \frac{\alpha}{p} + an(p). \quad (6)$$

Thus a higher housing cost or a tighter access constraint reduces fertility. Young housing demand is adult housing, α/p , plus the space required by children, $an(p)$.

This is a stripped-down version of the household mechanism already in the paper. There, the other-goods cost of children is χ , the space required per child is κ , and the relevant housing cost depends on tenure and on whether the household is constrained. In the notation of the current model, the simple expression $q + a(p + \omega)$ corresponds to $\chi + \kappa(r^m + \zeta_i^m)$: ω summarizes the shadow cost of limited housing access. The simple model removes heterogeneity, saving, and the rent-versus-own decision only to make the demographic feedback transparent.

2.3 Equilibrium and the steady state

Housing clearing adds young and old demand:

$$\bar{H}p_t^\varepsilon = Y_t h_Y(p_t) + O_t h_O(p_t). \quad (7)$$

For inherited cohort masses, the left side rises and the right side falls with p_t , so the clearing price is unique.

Children become the next young cohort, while the current young become old:

$$Y_{t+1} = \frac{n(p_t)}{\bar{n}} Y_t, \quad O_{t+1} = Y_t. \quad (8)$$

The ratio $n(p_t)/\bar{n}$ is the number of next-period young households produced by each current young household.

A steady state has a constant housing cost and constant positive cohort masses, so $n(p^*) = \bar{n}$ and $Y^* = O^*$.

Proposition 1 (Steady state). *A unique positive steady state exists if and only if $\theta > \bar{n}(q + a\omega)$. It is*

$$p^* = \frac{\theta/\bar{n} - q}{a} - \omega, \quad Y^* = O^* = \frac{\bar{H}(p^*)^{\varepsilon+1}}{\alpha + \bar{h}_O + a\bar{n}p^*}. \quad (9)$$

A lower value of children θ or a higher access cost ω lowers both the steady-state housing cost and population. A higher $\bar{h}_O$ reduces population but does not change the steady-state housing cost. A proportional increase in housing supply $\bar{H}$ raises population in the same proportion and also leaves the steady-state housing cost unchanged.

The cohort law first determines the housing cost consistent with replacement, $n(p^*) = \bar{n}$. Housing clearing then determines how many households can live in the market at that cost. This is why a permanent increase in housing supply raises population rather than lowering the steady-state housing cost.

Even if market housing were free, a child would still cost $q + a\omega$. Hence cheaper housing cannot restore replacement when $\theta \leq \bar{n}(q + a\omega)$.

2.4 Comparative statics

Suppose the economy begins in a steady state and a permanent reform arrives after Y_t and O_t have been inherited. The impact and steady-state effects are:

	Impact p_t	Impact n_t	Long-run p^*	Long-run n^*	Long-run population
Lower access cost, $\omega \downarrow$	+	+	+	0	+
Lower old housing demand, $\bar{h}_O \downarrow$	−	+	0	0	+
Greater housing supply, $\bar{H} \uparrow$	−	+	0	0	+

Lowering ω makes family housing easier to obtain. Fertility and housing demand rise immediately, so part of the reform appears as a higher market price. Lower old housing demand and greater supply instead reduce the price on impact. In the new steady state, fertility is again $\bar{n}$. For the latter two reforms, the housing cost returns to its original level and the additional housing supports a larger population.

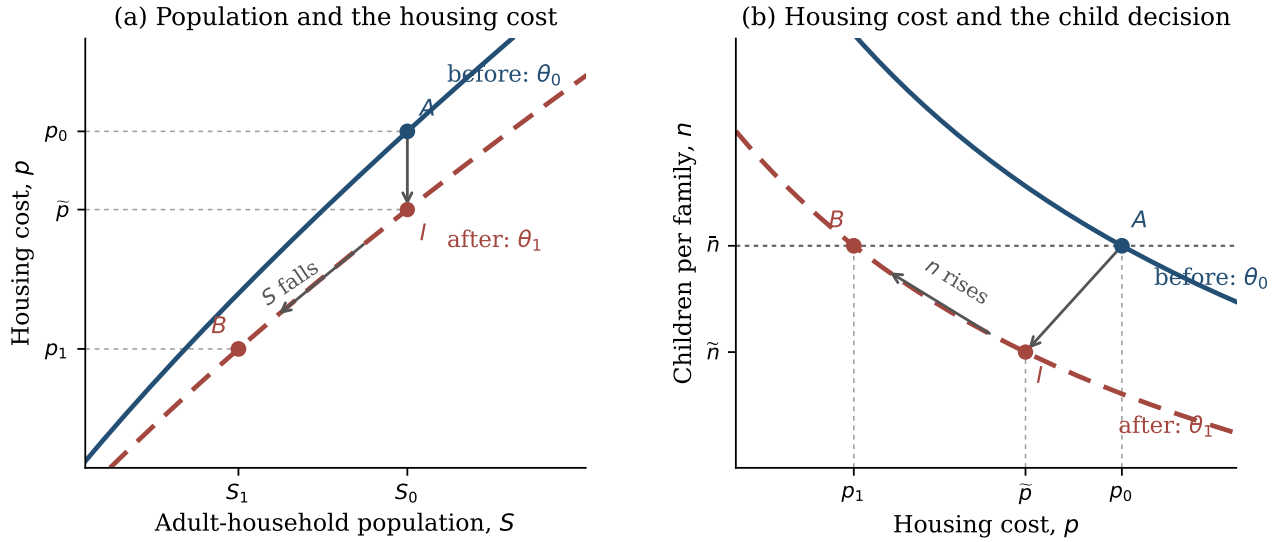


Figure 1: The child decision determines the steady-state housing cost through $n(p^*) = \bar{n}$. Housing clearing then determines population at that cost. A fall in θ initially puts fertility below replacement; as population falls, housing becomes cheaper and partly offsets the initial decline. The solid blue housing-clearing schedule applies before the fall in θ and the dashed red schedule after it; A, I, and B denote the initial steady state, impact allocation, and new steady state.

2.5 Demographic momentum

Let $\mu_t = O_t/Y_t$ be the number of old households per young household, and let $m_t = n(p_t)/\bar{n}$ be the number of next-period young households produced by each current young household. When fertility is below replacement, $m_t < 1$, total household population still rises if

$$\mu_t < m_t < 1. \quad (10)$$

For example, 100 young and 70 old households become 80 young and 100 old households when $m_t = 0.8$. Population rises from 170 to 180 even though the young cohort is not replacing itself.

If preferences and housing supply do not change again between t and $t + 1$, the housing cost also rises when

$$(1 - \mu_t)h_O(p_t) > (1 - m_t)h_Y(p_t). \quad (11)$$

The smaller young cohort reduces demand by $(1 - m_t)Y_t h_Y(p_t)$. At the same time, the current young cohort replaces a smaller old cohort and raises old demand by $(1 - \mu_t)Y_t h_O(p_t)$. The price rises when the second change is larger.

Starting from an exact steady state, $\mu_t = 1$. In this closed two-cohort model, a fertility decline lowers the housing cost on impact and lowers population in the next period. A period in which fertility is below replacement while population and housing costs rise therefore requires an inherited cohort imbalance, produced by earlier fertility, immigration, or both.

On impact, the fertility decline lowers the housing cost relative to the no-shock allocation. Demographic momentum can nevertheless make the realized cost rise over time. The sign comparison above does not by itself order the entire shocked and no-shock price paths.

2.6 Intergenerational policy

Lowering ω directly improves young households' access to family space. Lowering $\bar{h}_O$ instead releases housing from old households and reduces the market price during the transition. Both policies raise fertility, but through different margins. A general expansion of supply also lowers the transition price, without changing which generation initially releases the housing.

With a fixed adult population, reallocating a home toward a constrained young household affects current tenure and fertility. With the cohort law in equation (8), the resulting birth also changes the number of future entrants. That second effect is lost when population is renormalized after every solution.

3 Quantitative implementation

3.1 The household model is unchanged

The standing quantitative model has a one-shot completed-family-size choice. Renters face a size limit, buyers face a down-payment constraint, owners choose from a housing ladder, and selling costs and bequest motives make older owners reluctant to release housing. The population extension changes none of these objects. It only adds an outer equilibrium calculation around the household model.

At any candidate asset price P , the existing solver returns four normalized objects: required entrants $E(P)$, mature locally born households $B(P)$, housing demand per adult household $D(P)$, and the complete set of household choices. Let $H^s(P)$ denote the existing housing-supply curve. For fixed outside inflow M and local retention ρ , stationary population is

$$S(P) = \frac{M}{E(P) - \rho B(P)}. \quad (12)$$

This open steady state is finite only when $E(P) > \rho B(P)$. The quantitative outer loop chooses P so that

$$S(P)D(P) = H^s(P). \quad (13)$$

This uses the old household solver as a black box. There are no new household states and no recalibration. In a closed market with full local retention, $M = 0$ and $\rho = 1$, and a positive steady state instead requires

$$B(P^*) = E(P^*), \quad S^* = \frac{H^s(P^*)}{D(P^*)}. \quad (14)$$

Thus the closed calculation first finds the price that restores replacement and then uses housing clearing to determine population, exactly as in the two-generation model.

3.2 What the saved one-shot estimate implies

This is a fixed-parameter mechanism calculation, not a promoted calibration. The first-child housing response used in that saved estimate has since been withdrawn, and its replacement remains under empirical review.

At the saved circulated one-shot estimate, the four-year entrant flow is $E_0 = 0.061733$ per adult household and mature locally born households are $B_0 = 0.060701$. Hence

$$\frac{B_0}{E_0} = 0.983278. \quad (15)$$

With full local retention, the normalized population is therefore an open steady state if outside inflow is $M_0 = E_0 - B_0 = 0.001032$ per adult household every four years. This is 1.67 percent of required entrants. More generally, a larger outside share can nest the same benchmark with proportionately lower local retention. Neither object was estimated in that exercise, so the policy calculations report their sensitivity.

The closed root is far from the saved point. Its asset price is $P_0 = 1.0465$, while $B(P) = E(P)$ at $P^* = 0.4797$, or 45.84 percent of the saved price. The completed-fertility equivalent at the root is 2.120. Under the maintained static supply curve, housing clearing implies $S^* = 0.167$ relative to the normalized population; holding the current housing stock fixed instead gives $S^* = 0.653$. These are steady-state endpoint calculations, not forecasts. Their main message is that treating the United States as closed while retaining the current fertility schedule is a very strong quantitative change, not a harmless normalization.

Figure 2 shows the two conditions computed from the full household model. Panel A finds the price at which locally born households replace exiting households. Panel B holds the outside inflow fixed at its benchmark value and finds population in two ways: from renewal and from housing-market clearing. Their intersection reproduces the saved estimate.

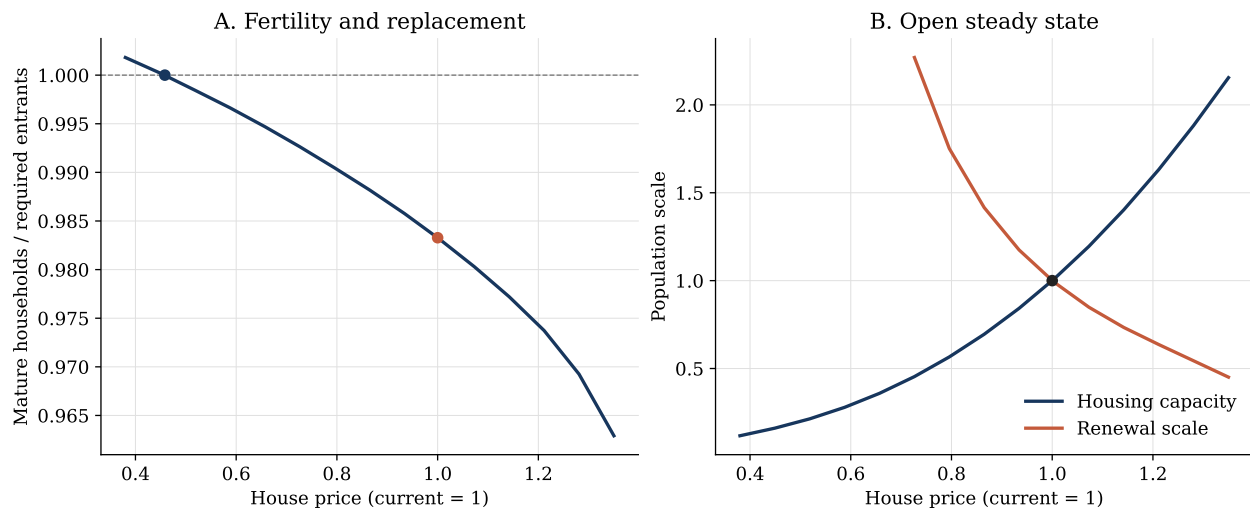


Figure 2: Stationary closure in the full household model. Panel A plots mature locally born households relative to required entrants. Panel B plots the adult household mass permitted by housing supply and the mass implied by the fixed outside inflow. Price is divided by its value at the saved estimate, and adult household mass is normalized to one there.

3.3 How the closure changes the paper's policy experiments

We rerun the paper's fiscally balanced policies without changing their rules. The baseline levies a 1 percent annual property tax on all occupied housing and rebates the revenue equally. The purchase-grant policy uses part of that revenue for a 0.4 model-unit grant on a birth-state renter-to-owner purchase of

six or more rooms. The final policy raises the annual property tax to 2 percent and keeps the same grant; all remaining revenue is rebated. Estimated parameters are held fixed throughout, so the rebated-tax baseline is a counterfactual reference rather than a recalibrated benchmark.

The table compares the old fixed-population calculation with an open steady state in which outside entrants are 20 percent of the baseline entrant flow. Local retention is chosen so that the open closure reproduces exactly the same baseline population and price. The resulting M and ρ are then held fixed across policies. Population, denoted S , is adult household mass. Total births are the four-year birth flow across all adult households, so they combine births per household with the change in the number of households.

Table 1: Policy effects with fixed and endogenous population

	Fixed population		Renewal: 20% baseline outside share		
	ΔP	Δ births/adult	ΔP	ΔS	Δ total births
Purchase grant	+0.01	+0.20	+0.32	+0.75	+0.94
2% tax + purchase grant	-17.86	+0.56	-17.10	+2.24	+2.79

Notes: Entries are percent changes from the fiscally balanced 1 percent property-tax baseline.

With a 20 percent outside share, the main fixed-population conclusions survive. The purchase grant has almost no price effect, while the tax-and-grant package substantially lowers the house price. The new closure adds a population margin: the two policies raise total births by 0.94 and 2.79 percent, compared with increases in births per household of 0.20 and 0.56 percent under fixed population.

The size of this margin depends sharply on openness. With full local retention, the funded baseline needs outside inflow equal to only 0.89 percent of required entrants. In that nearly closed case, the purchase grant raises population by 10.33 percent and the tax-and-grant package raises it by 39.14 percent. The larger population absorbs housing, so the price effects become +4.12 and -5.88 percent. These are sensitivity results, not estimates: outside inflow and local retention were not calibrated. Figure 3 shows the contrast.

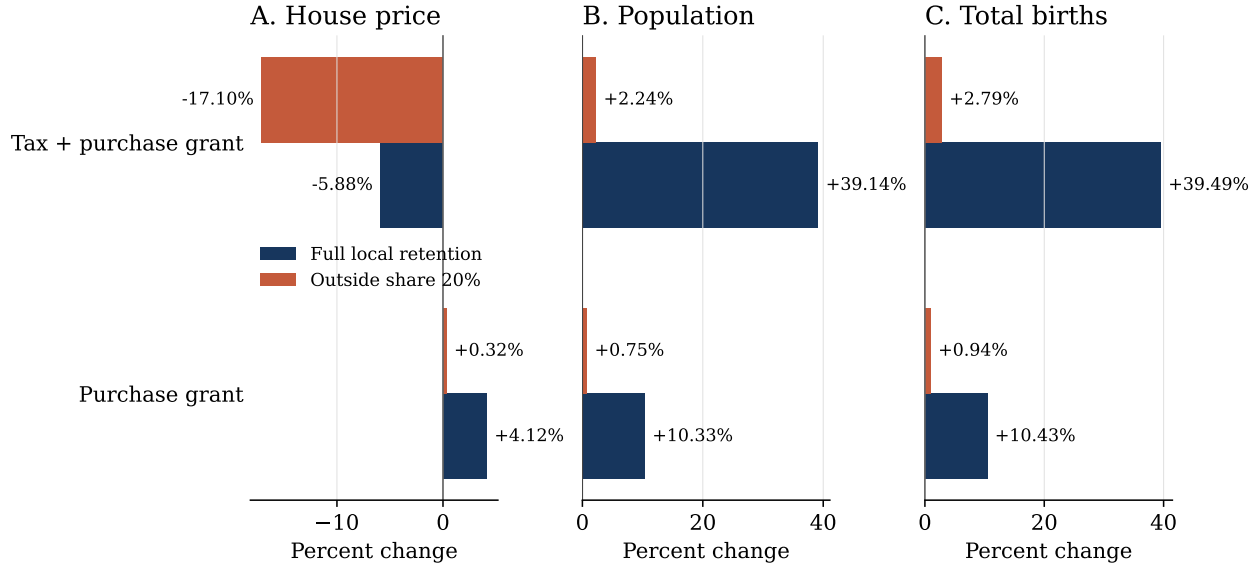


Figure 3: The paper’s funded policies under the stationary population closure. Each bar is a percent change from the fiscally balanced 1 percent property-tax baseline. “Full local retention” still includes the minimum outside inflow needed to reproduce that baseline. The 20 percent case fixes M and ρ at the more open values that nest the same baseline.

This exercise isolates steady-state incidence. It does not establish how fast the economy moves between the two steady states, and it is not a welfare calculation.

3.4 Why the historical transition is not yet a result

The stationary forward accounting can carry unnormalized mass and nests the existing solution to machine precision. The current state space cannot yet initialize a U.S. transition consistently. In our 2023 ACS large-metro household-head sample, ages 18–21 are 0.94 percent of heads aged 18–85. The model instead puts every new household in its age-18 state, with an entrant flow equal to 6.17 percent of adult household mass. Combining the two objects would multiply the youngest cell by more than six and create an artificial population and price boom.

That jump is household formation, not demographic momentum. The model has no non-head adult state or formation delay. Reweighting by all adults reduces the jump but mixes persons with household wealth, tenure, and housing demand. We therefore do not report either path as a U.S. transition.

The first empirical bridge is a household-formation pipeline that maps mature children into new households by age. The level distribution could then be propagated without normalization. A perfect-foresight asset-price path would be a further, separate requirement.

3.5 The remaining empirical object

The long-run feedback also depends on the response of fertility to housing costs and access; fertility levels at one price do not identify that slope. The useful evidence therefore moves housing costs or credit access and measures fertility separately for renters, prospective buyers, and incumbent owners. Until that response and household formation are disciplined, the steady-state incidence calculations remain provisional.

4 Conclusion

The proposed change is narrow: an outer loop uses the existing model’s entrant, child-maturation, and housing-demand objects to determine population and the housing price jointly. The household problem and intergenerational mechanism remain intact.

Fertility determines the price consistent with replacement, while housing clearing determines population at that price. The saved one-shot estimate is an open steady state and lies far from its implied closed steady state. With a materially open market, endogenous population preserves the policy ranking but raises total births more than births per household. Near the closed boundary, the same policies instead produce large population responses and smaller price responses. The next step is to measure household formation well enough to connect these steady states with a defensible transition.

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